

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 03-08-25
Time: 3 Hrs Max. Marks: 186

03-08-25_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-1_SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

Rankers Academy JEE
THE NARAYANA GROUP

For More Join: @IITJEE_Advanced

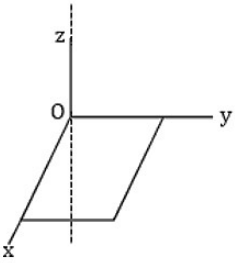
PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

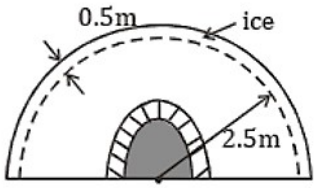
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A square of mass m is made of thin wire with sides of length a. Calculate the rotational inertia about the z-axis.



(A) $\frac{7ma^2}{6}$ (B) $\frac{5ma^2}{6}$ (C) $\frac{7ma^2}{12}$ (D) $\frac{5ma^2}{12}$

2. An igloo, a hemispherical enclosure built of ice (k = 1.67 J/m-s°C), has an inner radius of 2.50 m. The thickness of the ice is 0.5 m. At what rate must thermal energy be generated to maintain the air inside the igloo at 5°C when the outside temperature is –40°C? Ignore all thermal energy losses through the ground or by air currents.



(A)2.25 πkW (B) 4.5 πkW (C) 1.25 πkW (D) 5 πkW

3. Assume temperature of 27°C, p = 10^5Pa, n = 1 moles, diameter of air molecules as 10^{-10} m, which of the following is correct?

(A) Mean free path > interatomic spacing

(B) Mean free path = interatomic spacing

(C) Mean free path < interatomic spacing

(D) Data insufficient

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4. If we want that a simple pendulum oscillates with same time period in summer as it did in winter, what changes to the simple pendulum should we do during summer?

(A) Increase the length (B) Decrease the length

(C) Increase mass of bob (D) Decrease mass of bob

5. A car is moving towards a stationary observer at a speed of 108 km/hr. The speed of sound is 330 m/s. If the observer hears the frequency of horn as 550 Hz, what frequency will he hear after the car has passed?

(A) 550 Hz (B) 500 Hz (C) 458.33 Hz (D) 421.6 Hz

6. Which of the following is a CORRECT statement?

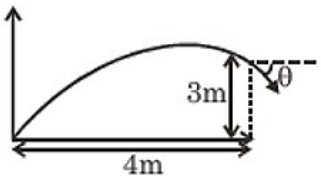
(A) $2.3056 + 10.138 - 7.4671 = 4.9765$ (B) $2.38 \times 1.0 = 2.38$

(C) $\frac{8.05}{3.1} = 2.6$ (D) $(1.11 - 0.1) \times 9.0 = 9.0$

7. An electron in a research apparatus follows a circular path. During the electron's first revolution in the apparatus, its speed is v_0 and the radius of its circular path is R. Each time it completes one revolution, it receives a "kick" and gains an additional speed of $\frac{v_0}{100}$. The electron follows a circular path such that the magnitude of its acceleration is always the same. What is the radius of the circular path after the electron has received 10 kicks?

(A) $\frac{R}{1.1}$ (B) 1.1R (C) 1.21R (D) $\frac{R}{1.21}$

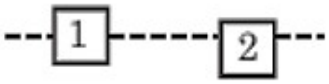
8. A projectile travels with its velocity directed at an angle θ below horizontal, 1 sec after projection as shown. What is the value of θ?



(A) $\tan^{-1}(2)$ (B) $\tan^{-1}\left(\frac{1}{2}\right)$ (C) 37° (D) 53°

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9. Two blocks of same mass but different density ρ_1 and ρ_2 are floating in water. Block-2 has higher fraction of its volume submerged in water. If buoyant force on them are B_1 and B_2



(A) $B_1 > B_2, \rho_1 > \rho_2$ (B) $B_1 < B_2, \rho_1 < \rho_2$

(C) $B_1 = B_2, \rho_1 > \rho_2$ (D) $B_1 = B_2, \rho_1 < \rho_2$

10. In half deflection method, an ideal cell, a resistor R =10,000 Ω, and galvanometer are connected in series. When the galvanometer is shunted with 10 Ω resistance, its reading drops to half the initial value. The resistance of galvanometer is

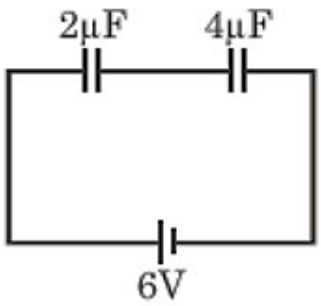
(A) 5 Ω (B) 10 Ω (C) 20 Ω (D) 25 Ω

11. A plastic ball is rising in water with terminal speed v. If we use a different plastic whose density is double that of the original ball but still less than density of water, keeping the radius same, the terminal speed would be

(A) Double (B) Half

(C) More than original value (D) Less than original value

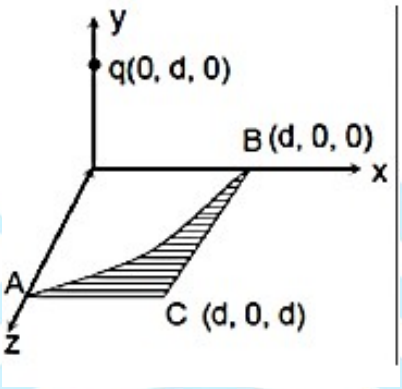
12. Two air filled capacitors are connected as shown. The left capacitor is suddenly filled with a dielectric of constant K. The work done by battery after that is 24 μJ. Dielectric constant K is



(A) 2 (B) 3 (C) 4 (D) 6

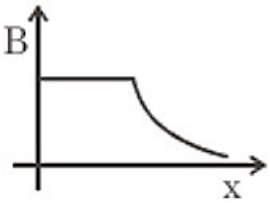
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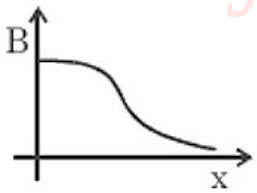
13. A point charge $q = 8.85 \times 10^{-9}$ C is placed at (0, d, 0) where d = 1 m. An insulating surface ABCA is on the x-z plane, which is bounded by a quarter circle AB and two equal length straight line BC and CA as shown in the figure. Electric flux through the surface ABCA in Nm^2/C is (nearest integer) Take $\varepsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{Nm^2}$



(A) 10 (B) 5 (C) 15 (D) 25

14. An infinite cylindrical current carrying wire carries a current that is uniformly distributed over its cross section. The magnetic field as a function of distance from it's axis is

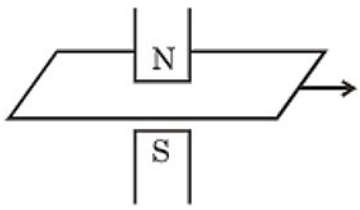
(A) 

(B) 

(C) 

(D) 

15. A metal sheet is being pulled out of a region of magnetic field. If the sheet is non-magnetic?



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(A) No force is required and no heat is dissipated in sheet

(B) Force is required but no heat is dissipated in sheet

(C) No force is required but heat is dissipated in the sheet

(D) Force is required and heat is dissipated in the sheet

16. Atomic mass of Bromine, Selenium and Krypton are given as $M_{Br}^{80} = 79.918528\,u$, $M(^{80}Kr) = 79.916376\,u$, $M(^{80}Se) = 79.916521\,u$. Which of the following β decays are possible?

(i) ${}^{80}_{35}Br \rightarrow {}^{80}_{36}Kr + {}^{0}_{-1}e + \bar{\nu}$ (ii) ${}^{80}_{35}Br \rightarrow {}^{80}_{34}Kr + {}^{0}_{+1}e + \nu$ (iii) ${}^{80}_{35}Br + {}^{0}_{-1}e \rightarrow {}^{80}_{34}Se + \nu$

(A) all 3 (B) (i) and (iii) only

(C) (ii) and (iii) only (D) (i) and (ii) only

17. If a system of two atoms collide inelastically,

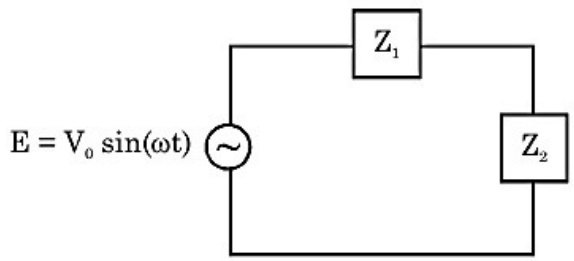
A) Kinetic energy of the system remains the same.

B) Heat is produced

C) Sound is produced

D) At least one of the atoms undergoes excitation.

18. The impedance Z_1 in figure can be regarded as a pure resistance $R_1 = 12\ \Omega$, whereas the impedance Z_2 is associated with a series resistance $R_2 = 8\Omega$ and a capacitance $C = 1\mu F$. If $f = 5000$ Hz and $V_0 = 30V$, what is the power dissipated in Z_2 (in W)? Take $\pi^2 = 10$.



(A) 2.57 (B) 3.12 (C) 257 (D) Zero

For More Join: @IITJEE_Advanced

19. N-division on the main scale of a vernier callipers coincide with (N + 1) divisions on the vernier scale. If each division on the main scale is of a units, determine the least count (L.C.) of the instrument

(A) $\left(\frac{a}{N+1}\right)$unit (B) $\left(\frac{2a}{N+1}\right)$unit

(C) $\left(\frac{a}{2(N+1)}\right)$unit (D) $\left(\frac{3a}{N+1}\right)$unit

20. High-power lasers are used to compress gas plasmas by radiation pressure. The reflectivity of a plasma is unity(equal to 1) if the electron density is high enough. A laser generating pulses of radiation of power 1.5 GW is focused onto 1.3 mm^2 of high-electron-density plasma. Find the pressure exerted on the plasma

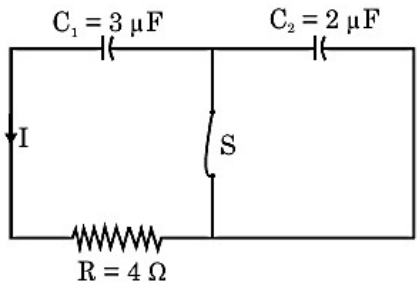
(A) 4.2 MPa (B) 1 MPa (C) 7.7 MPa (D) 10 MPa

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

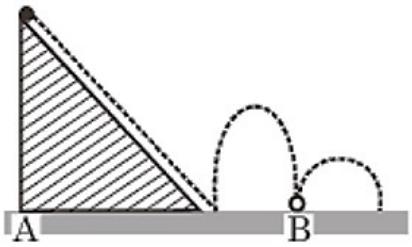
21. A wave given by equation y = 1 mm sin $\left(\frac{\pi x}{30} - 5\pi t\right)$ is produced in a string 100 m long of mass 1 kg. What is the tension (in N) in string? x is in m and t is sec.

22. A charged capacitor (C_1 = 3 μF) is getting discharged in the circuit shown and the current I was observed to be 2.5A. Now, switch 'S' is opened. Determine the amount of heat (in μJ) that will be liberated in the circuit after 'S' is opened.



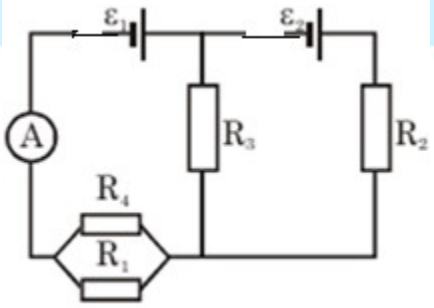
For More Join: @IITJEE_Advanced

23. A small elastic ball of mass m is placed at the apex of a 45° inclined plane as shown in the figure below. The ball is allowed to slip without friction down the plane (along the dotted line), hit the ground (as shown) and bound along it. If the height of the inclined plane is h and the coefficient of restitution between the ball and the ground is 0.5, then the distance AB, as marked on the figure, will be N × h, where N is



24. The velocity of liquid (v) in steady flow at a location through cylindrical pipe is given by $v = v_0\left(1-\frac{r^2}{R^2}\right)$, where r is the radial distance of that location from the axis of the pipe and R is the inner radius of pipe. If R = 10 cm, volume rate of flow through the pipe is $\frac{\pi}{2}\times10^{-2}\ m^3s^{-1}$ and the coefficient of viscosity of the liquid is 0.75 N sm^{-2}, find the magnitude of the viscous force per unit area, in Nm^{-2} at r = 4 cm.

25. In the circuit in figure we have $\varepsilon_1 = \varepsilon_2 = 100V$, $R_1 = 20\Omega$, $R_2 = 10\Omega$, $R_3 = 40\Omega$ and $R_4 = 30\Omega$. Find the reading of the ammeter (in A). Disregard the resistance of the battery and the ammeter.



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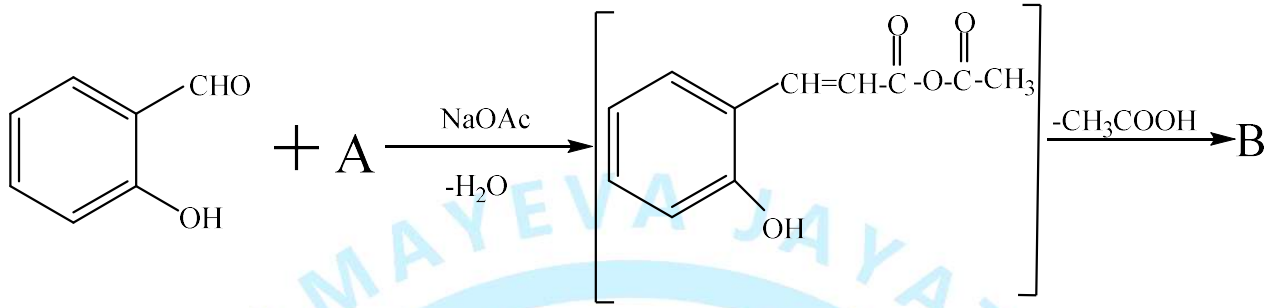
CHEMISTRY **MAX.MARKS: 100**

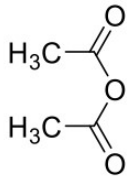
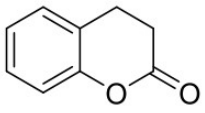
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

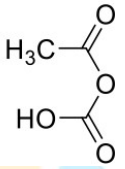
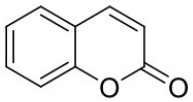
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

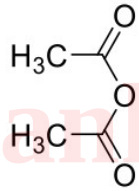
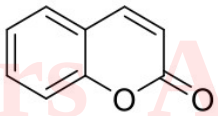
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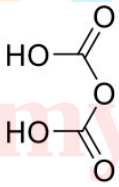
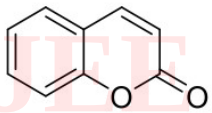
26. Give the missing product B and the reagent A in the reaction scheme below



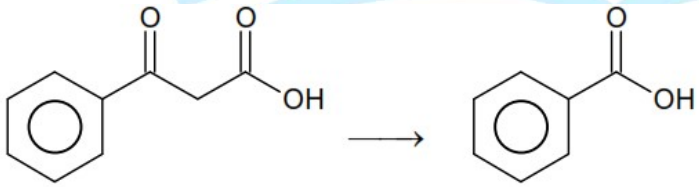
(A) **A:**  **B:** 

(B) **A:**  **B:** 

(C) **A:**  **B:** 

(D) **A:**  **B:** 

27. The following conversion reaction CANNOT be carried out by using reaction sequences:



(A) $\xrightarrow{Zn-Hg/HCl} \xrightarrow{Br_2, hv} \xrightarrow{KCN} \xrightarrow{H_3^{\oplus}O, \Delta}$

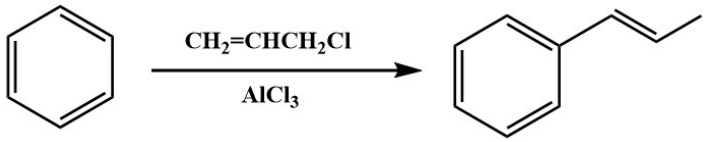
(B) $\xrightarrow{NaBH_4} \xrightarrow{Al_2O_3, \Delta} \xrightarrow{O_3/H_2O(Oxidation)}$

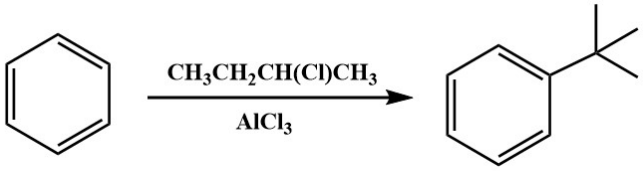
(C) $\xrightarrow{\Delta} \xrightarrow{I_2 + NaOH} \xrightarrow{H^{\oplus}}$

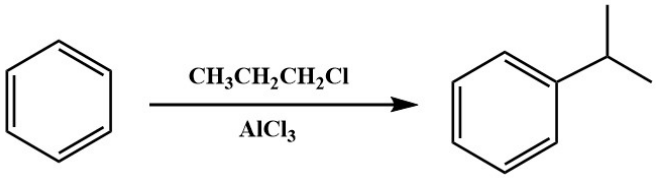
(D) $KMnO_4 / HO^- / \Delta$

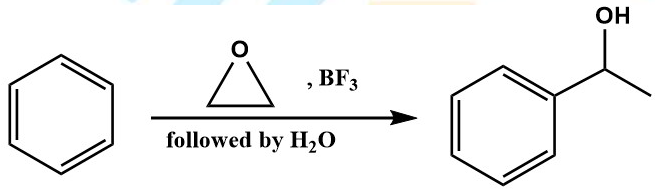
For More Join: @IITJEE_Advanced

28. Which reaction is shown with the main product of the Friedel-Crafts alkylation reactions given below?

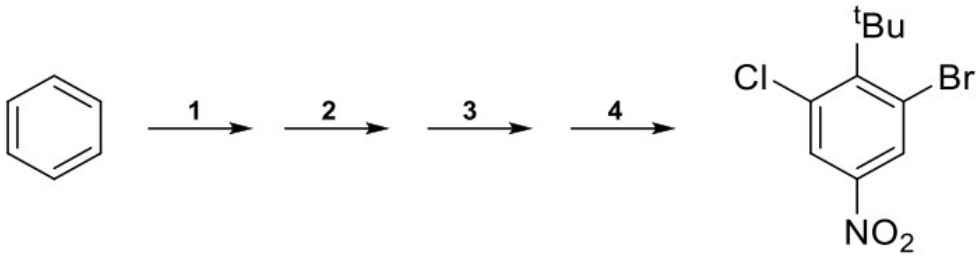
(A) 

(B) 

(C) 

(D) 

29. Give the correct order of the reaction steps:



(A) (1) $(CH_3)_3CCl$, $AlCl_3$; (2) HNO_3, H_2SO_4; (3) Br_2, $AlBr_3$; (4) Cl_2, $AlCl_3$

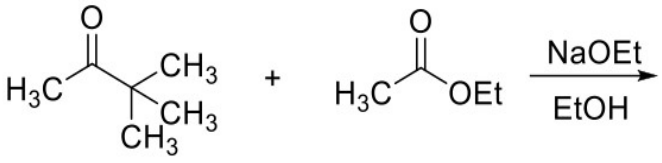
(B) (1) Br_2, $AlBr_3$; (2) Cl_2, $AlCl_3$; (3) $(CH_3)_3CCl$, $AlCl_3$; (4) HNO_3, H_2SO_4

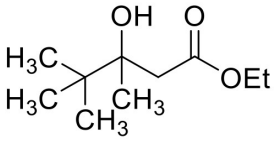
(C) (1) Br_2, $AlBr_3$; (2) Cl_2, $AlCl_3$; (3) HNO_3, H_2SO_4; (4) $(CH_3)_3CCl$, $AlCl_3$

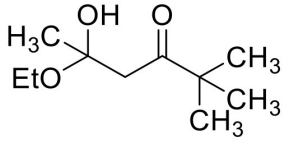
(D) (1) $(CH_3)_3CCl$, $AlCl_3$; (2) HNO_3, H_2SO_4; (3) NBS; (4) Cl_2, $AlCl_3$

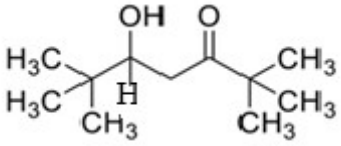
For More Join: @IITJEE_Advanced

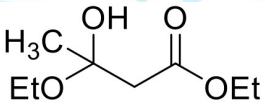
30. What will be the main addition product of the first step of the following reaction?



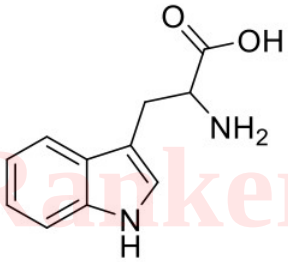
(A) 

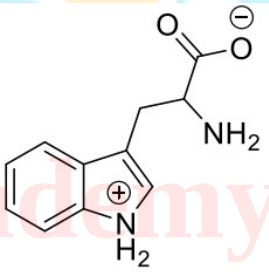
(B) 

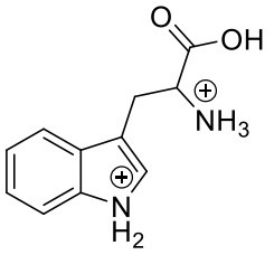
(C) 

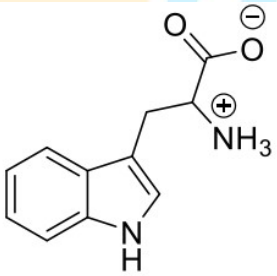
(D) 

31. What is the predominant structure of tryptophan at pH = 7?

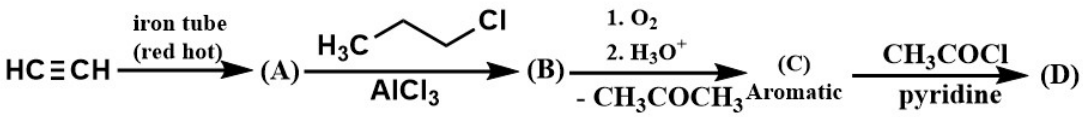
(A) 

(B) 

(C) 

(D) 

32. The Compound D is



(A) Phenyl Ethanoate

(B) Methyl Benzoate

(C) Benzophenone

(D) Ethyl Benzoate

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33. The wavelength of light emitted by $He^{\oplus}$ ion and unknown single electron species, is 16x unit and 4x unit respectively for transition between similar orbits. Basis this, the unknown single electron species can be

(A) H-atom (B) Li^{2+} ion (C) Be^{3+} ion (D) B^{4+} ion

34. Which of the following must be true for a spontaneous exothermic process?

(A) Only that $\Delta S_{sys} > 0$

(B) Only $\Delta S_{sys} < 0$ and the magnitude of ΔS_{sys} < the magnitude of ΔS_{surr}

(C) Only $\Delta S_{sys} < 0$ and the magnitude of ΔS_{sys} > the magnitude of ΔS_{surr}

(D) Either $\Delta S_{sys} > 0$, or else $\Delta S_{sys} < 0$ but the magnitude of ΔS_{sys} < the magnitude of ΔS_{surr}

35. For a saturated aqueous solution of LiF · Ksp = 2.0×10^{-12} if Ka of HF = 1.0×10^{-4}. Ignoring the hydrolysis of Li^{+}, when pH = 4 for given solution, the solubility of LiF at 25°C

(A) 2.0×10^{-12} M (B) 4×10^{-12} M

(C) $2 \times 10^{-6} M$ (D) 2.5×10^{-10} M

36. The reaction A(g) + 2B(g) → C(g) + D(g) is an elementary process. In an experiment involving this reaction, the initial partial pressures of A and B are $p_A = 0.60$ atm and $p_B = 0.80$ atm respectively. When $p_C = 0.20$ atm, the rate of the reaction relative to the initial rate is

(A) $\frac{1}{6}$ (B) $\frac{1}{12}$ (C) $\frac{1}{36}$ (D) $\frac{1}{18}$

37. Plot of $\frac{1}{x_A}$ Vs $\frac{1}{Y_A}$ (x_A mole fraction of A in liquid state and Y_A in vapour state) is linear whose slope and intercept respectively are given

(A) $p_B^0 / p_A^0, \frac{p_B^0 - p_A^0}{p_B^0}$ (B) $p_A^0 - p_B^0, \frac{p_A^0 - p_B^0}{p_B^0}$

(C) $p_A^0 / P_B^0, \frac{p_B^0 - p_A^0}{p_B^0}$ (D) $p_B^0 / p_A^0, \frac{p_A^0 - p_B^0}{p_B^0}$

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38. The degree of dissociation (α) of PCl_5 obeying the equilibrium $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ is related to the equilibrium pressure by (when α is very small with respect to 1)

(A) $\alpha \propto \frac{1}{P^4}$ (B) $\alpha \propto \frac{1}{\sqrt{P}}$ (C) $\alpha \propto \frac{1}{P^2}$ (D) $\alpha \propto P$

39. HA is a weak monobasic acid. Molar conductivity of aqueous solution of HA is 250 S $cm^2 mol^{-1}$ and pH of the solution is maintained at 5. What is the Ka of HA of 298 K .

Given:

$\Lambda_m^0(NaA) = 150\ Scm^2\ mol^{-1}$, $\Lambda_m^0(HCl) = 425\ Scm^2\ mol^{-1}$, $\Lambda_m^0(NaCl) = 125\ Scm^2\ mol^{-1}$

(A) 1.25×10^{-5} (B) 3.68×10^{-4} (C) 4.52×10^{-5} (D) 2.63×10^{-3}

40. Which is incorrect for the property mentioned

(A) $Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$ (*Basic strength*)

(B) LiF < NaF > KF > RbF > CsF (Lattice energy)

(C) $CuS < FeS < ZnS$ (Solubility in water)

(D) $BeCO_3 > MgCO_3 > CaCO_3 > BaCO_3$ (Thermal stability)

41. The correct statement about ICl_5 and ICl_4^- is

(A) ICl_5 is square pyramidal and ICl_4^- is tetrahedral

(B) ICl_5 is square pyramidal and ICl_4^- is square planar

(C) Both are isostructural

(D) ICl_5 is trigonal bipyramidal and ICl_4^- is tetrahedral

42. Which set of elements have electronegativity value more than that of carbon

(A) Cl, Br, I, S (B) I, S, P, As (C) Br, I, S, P (D) I, P, As, Si

43. $Na_2S_2O_3 + I_2 \longrightarrow A + B$, $A + KMnO_4(aq) \xrightarrow{\text{mild alkaline}} \underset{(\text{soluble})}{C} + \underset{\text{Black brown ppt}}{D\downarrow}$

$D + HCl \longrightarrow MnCl_2 + E(gas) + H_2O$. Which is correct for A, C and E

(A) NaI, I_2, ClO_2 (B) NaI, $IO_3^{\ominus}$, Cl_2

(C) Na_2S, SO_2, H_2S (D) Na_2S, $IO_3^{\ominus}$, Cl_2

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44. Consider a salt AB

$\text{Cation(aq)} + H_2S \xrightarrow{HCl} \text{Black ppt}$

$AB(s) + NH_4OH \longrightarrow \text{salt dissolve as complex}$

$\text{anion} + \underset{\text{conc.}}{H_2SO_4} \longrightarrow \text{Brown fumes formed}$

$AB(s) + Na_2S_2O_3(aq) \longrightarrow \text{salt dissolve as complex}$

(i) $AgNO_2$ (ii) AgBr (iii) $CuBr_2$ (iv) $Hg(NO_2)_2$

Salt AB can be

(A) (i) only (B) (i) and (ii) only

(C) (i), (ii) and (iii) only (D) (i), (ii), (iii) and (iv)

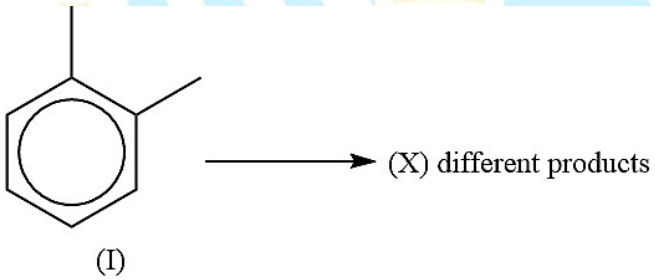
45. The total possible isomer of $\left[Co(H_2O)_5(NO_2)\right](NO_3)_2$

(A) 6 (B) 8 (C) 10 (D) 9

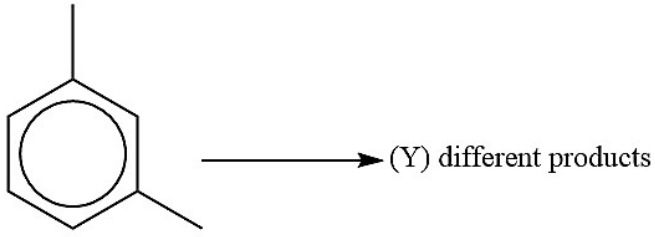
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Identify the sum of total number of different products (X) and (Y) that are formed when below compounds (I) and (II) are separately reacted with ($\xrightarrow[Zn/H_2O]{O_3/heat}$)?

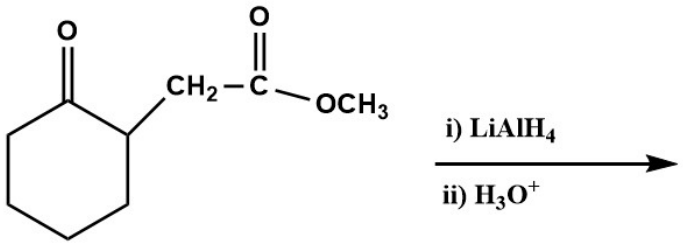


(I)



(II)

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47. 

When one mole of substrate is taken, what is the volume of the gas liberated at STP when the products are treated with excess sodium metal. (in litres) (GMV = 22.40 lit)

48. The total pressure observed by mixing two liquids A and B is 350 mm Hg when their mole fractions are 0.7 and 0.3 respectively

The total pressure becomes 410 mm Hg if the mole fractions are changed to 0.2 and 0.8 respectively for A and B. The vapour pressure of pure A is _____ mm Hg. (Nearest integer) Consider the liquids and solutions behave ideally.

(check the calculation my answer is coming 310 and given is 314)

49. Number of total possible isomers of $[Co(dien)_2]Cl_3$ are _____[out of syllabus]

50. For the reaction $2N_2O_5 \longrightarrow 4NO_2 + O_2$. If initial pressure is 100 atm and rate constant of decomposition of N_2O_5 k is 3.38×10^{-5} s^{-1}, after 20 min the final pressure of N_2O_5 will be _____ atm. $[\ell n10 = 2.303 \, and \, e^{4.56544} = 96]$

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. The value of $\sum_{r=2}^{100} \frac{3^{r-1}(4-2r)}{r(r+1)(r-1)}$ is equal to

(A) $\frac{1}{2}-\frac{3^{100}}{100\times101}$ (B) $\frac{3}{2}-\frac{3^{101}}{101\times102}$ (C) $\frac{1}{2}-\frac{3^{101}}{101\times102}$ (D) $\frac{3}{2}-\frac{3^{100}}{100\times101}$

52. Number of positive integers n less than or equal to 1000 for which

$(\sin t + i\cos t)^n = \sin nt + i\cos nt$ holds true for all real t (where, $i=\sqrt{-1}$) is

(A) 245 (B) 250 (C) 300 (D) 350

53. Complete general solution of the equation $8\sin x\cos^5 x - 8\sin^5 x\cdot\cos x = \sqrt{3}$ is $(n \in Z)$

(A) $\frac{n\pi}{4}+(-1)^n\frac{\pi}{12}$ (B) $\frac{n\pi}{4}+(-1)^n\frac{\pi}{8}$

(C) $\frac{n\pi}{2}+(-1)^n\frac{\pi}{12}$ (D) $\frac{n\pi}{2}+(-1)^n\frac{\pi}{8}$

54. If $|x|+|y|+|z|=8$, x, y, z ∈ I, then the number of solutions of the equation is

(A) 258 (B) 111 (C) 195 (D) 129

55. Given $(1-x^3)^n=\sum_{k=0}^{n} a_k x^k (1-x)^{3n-2k}$ then the value of $3.a_{k-1}+a_k$ is _____

(A) ${}^nC_k\cdot 3^k$ (B) ${}^{n+1}C_{k-1}3^k$ (C) ${}^nC_{k-1}3^k$ (D) ${}^{n+1}C_k\cdot 3^k$

56. Total number of positive integral divisors of $n=3^5\cdot5^7\cdot7^9$ that are of the form 4k + 1 & $k\in W$ is equal to

(A) 240 (B) 120 (C) 60 (D) 30

57. If the real numbers x and y satisfy $(x+5)^2+(y-12)^2=14^2$, then the minimum value of x^2+y^2 is

(A) 2 (B) 1 (C) $\sqrt{2}$ (D) $\sqrt{3}$

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58. If $\begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\0 & 1\end{bmatrix}\begin{bmatrix}1 & 3\\0 & 1\end{bmatrix}\ldots\ldots\begin{bmatrix}1 & n\\0 & 1\end{bmatrix}=\begin{bmatrix}1 & 378\\0 & 1\end{bmatrix}$. Then n is equal to

(A) 27 (B) 26 (C) 376 (D) 378

59. Number of points of discontinuity of the function $f(x)=\sin(\{2^x+[2^x]+[3^{-x}]\})$ for $x \in [0, 2]$ is (where $[\cdot]$ and $\{\cdot\}$ represent the greatest integer function and fractional part function respectively)

(A) 15 (B) 4 (C) 3 (D) 16

60. A differentiable function $f: R^+ \to R$ satisfies the condition $f(x)-f(y)\geq \ln\left(\frac{x}{y}\right)+x-y$ for all $x, y \in R^+$. If $g(x) = f'(x)$, then the value of $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$ is

(A) 5050 (B) 5051 (C) 5151 (D) 5150

61. The minimum value of the function $f(x)=\frac{x^4+x^3+3x^2+x+1}{x^3+x}$, $x>0$ lies in the interval

(A) (2.8, 3.2) (B) (3.2, 3.4) (C) (3.4, 3.6) (D) (2.9, 3.4)

62. Let $f: R \to R$ be a non-constant function satisfying the relation $f(x+1)=\frac{1}{2}+\sqrt{f(x)-(f(x))^2}$ for all $x \in R$ and $\frac{1}{2}\leq f(x)\leq 1$. Then a period for all such f(x) is

(A) 2 (B) 3 (C) 1 (D) 5

63. If $f: R \to R$ is function satisfying the property $f(2x + 3) + f(2x + 7) = 2\ \forall x \in R$ and $f(1) = 4$, then $f(25)$ is equal to

(A) 23 (B) 4 (C) 80 (D) 8

64. If $A-\frac{I}{2}$ and $A+\frac{I}{2}$ both are orthogonal matrices, then

(A) A is a symmetric matrix of odd order

(B) A is a symmetric matrix of even order

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(C) A is a skew symmetric matrix of odd

(D) A is a skew symmetric matrix of even order

65. Four-unit squares are chosen at random on a chessboard. Then the probability that three of them are of one colour and other one is of opposite colour is

(A) $\frac{640}{1281}$ (B) $\frac{660}{1281}$ (C) $\frac{80}{427}$ (D) $\frac{160}{427}$

66. The sum of all the radii of the circles touching the co-ordinate axes and the line $3x + 4y = 12$, is

(A) 1 (B) 2 (C) 12 (D) $\frac{9}{2}$

67. A variable straight line with slope 4 intersects the hyperbola $xy = 1$ at two points P and Q. The locus of a point which divides PQ in the ratio 1 : 2 is

(A) $16x^2 - 10xy + y^2 = 2$ (B) $x^2 - 10xy + 16y^2 = 2$

(C) $16x^2 + 10xy + y^2 = 2$ (D) $x^2 + 10xy + 16y^2 = 2$

68. The value of $\int_0^{\pi} \frac{1-\cos 8x}{1-\cos x} dx$ is

(A) 4π (B) 8π (C) 16π (D) 64π

69. The shortest distance between the skew lines $\frac{x+3}{-4} = \frac{y-6}{3} = \frac{z}{2}$ and $\frac{x+2}{-4} = \frac{y}{1} = \frac{z-7}{1}$ is equal to

(A) 9 (B) 10 (C) 12 (D) 7

70. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{\sin x} dx$, $n \in N$, then the value of $\sum_{i=1}^{20} I_i$ is equal to

(A) 8π (B) 10π (C) 0 (D) π

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. The value of $72\left|\sum_{r=0}^{10}\cos^3\frac{\pi r}{3}\right|$ is equal to (where $|\cdot|$ represents modulus function) _____.

72. If one root of the equation $3x^2 + px + 3 = 0$, $(p > 0)$ is the square of the other, then p is equal to _____.

73. The local minimum value of $\left(x^2 + \frac{250}{x}\right)$ is λ, then $\frac{\lambda}{15}$ is equal to _____.

74. Let $\int_{-\pi/2}^{\pi/2} \sin^2 x \cdot \cos^2 x (\sin x + \cos x)\,dx = \frac{m}{n}$, where m, n are coprime positive integers, then (m + n) is equal to _____.

75. If the value of $\lim_{n\to\infty} \frac{n(1+2+3+...+n)(1^2+2^2+3^2+.....+n^2)(1^3+2^3+3^3+...+n^3)}{(1^4+2^4+3^4+....+n^4)^2}$ is $\frac{k}{p}$ (where k and p are coprime) then value of k + p is _____.

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